

Advanced Embedded Systems: Project Concepts

These advanced project ideas require students to move beyond high-level abstractions and engage with low-level architecture, RTOS, and communication protocols.

1. Custom RTOS Task Monitor

Goal: Monitor system performance and task execution in real-time.

Repo: <https://github.com/FreeRTOS/FreeRTOS-Kernel>

2. USB HID Device Emulator

Goal: Create a custom USB device using state machine logic.

Repo: <https://github.com/hathach/tinyusb>

3. Network-Enabled Sensor Hub

Goal: Implement a TCP/IP stack to serve sensor data via a web dashboard.

Repo: <https://github.com/lwip-tcpip/lwip>

4. Bare-Metal "Hello World"

Goal: Initialize a microcontroller from power-on without an OS.

Repo: <https://github.com/cpq/bare-metal-programming-guide>

5. Custom HAL Driver Development

Goal: Write a sensor driver using direct register-level access.

Repo: <https://github.com/libopencm3/libopencm3>